

Keyulu Xu

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EDUCATION	Massachusetts Institute of Technology Ph.D., Electrical Engineering and Computer Science • Advisor: Stefanie Jegelka • Research: deep learning, theory, optimization	Cambridge, MA <i>Sep. 2016 - Present</i>
	University of British Columbia B.Sc., Honours Mathematics and Computer Science • Advisor: Nick Harvey • Research: spectral graph theory, randomized algorithms	Vancouver, BC <i>Sep. 2012 - May. 2016</i>
POSITIONS	National Institute of Informatics <i>Visiting Researcher</i> • Hosted by Ken-ichi Kawarabayashi .	Tokyo, Japan <i>Feb. 2016 - Present</i>
	Two Sigma Investments <i>Quantitative Research Intern</i> • Alpha research. • Techniques (machine learning) team.	New York, NY <i>Jun. 2020 - Aug. 2020</i>
	Hudson River Trading <i>Fellow</i> • Research on high-frequency trading with deep learning. • HRT AI Labs.	New York, NY <i>Jun. 2019 - Aug. 2019</i>
	Google Inc. <i>Software Engineering Intern</i> • Distributed systems design and engineering.	New York, NY <i>May. 2015 - Aug. 2015</i>
PUBLICATIONS	Optimization of Graph Neural Networks: Implicit Acceleration by Skip Connections and More Depth Keyulu Xu, Mozhi Zhang, Stefanie Jegelka, Kenji Kawaguchi. <i>In Proceedings of the 38th International Conference on Machine Learning (ICML), 2021.</i>	
	Information Obfuscation of Graph Neural Networks Peiyuan Liao, Han Zhao, Keyulu Xu, Tommi Jaakkola, Geoffrey Gordon, Stefanie Jegelka, Ruslan Salakhutdinov. <i>In Proceedings of the 38th International Conference on Machine Learning (ICML), 2021.</i>	
	GraphNorm: A Principled Approach to Accelerating Graph Neural Network Training Tianle Cai, Shengjie Luo, Keyulu Xu, Di He, Tie-Yan Liu, Liwei Wang. <i>In Proceedings of the 38th International Conference on Machine Learning (ICML), 2021.</i>	
	How Neural Networks Extrapolate: From Feedforward to Graph Neural Networks Keyulu Xu, Mozhi Zhang, Jingling Li, Simon S. Du, Ken-ichi Kawarabayashi, Stefanie Jegelka. <i>In Proceedings of the 9th International Conference on Learning Representations (ICLR), 2021.</i> Oral Presentation (1.5% acceptance rate).	

What Can Neural Networks Reason About?

Keyulu Xu, Jingling Li, Mozhi Zhang, Simon S. Du, Ken-ichi Kawarabayashi, Stefanie Jegelka.
In Proceedings of the 8th International Conference on Learning Representations (ICLR), 2020.
Spotlight (4% acceptance rate).

Graph Neural Tangent Kernel: Fusing Graph Neural Networks with Graph Kernels.

Simon S. Du, Kangcheng Hou, Barnabás Póczos, Ruslan Salakhutdinov, Ruosong Wang, and Keyulu Xu.

Advances in Neural Information Processing Systems (NeurIPS), 2019.

Are Girls Neko or Shōjo? Cross-Lingual Alignment of Non-Isomorphic Embeddings with Iterative Normalization.

Mozhi Zhang, Keyulu Xu, Ken-ichi Kawarabayashi, Stefanie Jegelka, Jordan Boyd-Graber.

In Proceedings of the 57th Association for Computational Linguistics (ACL), 2019.

How Powerful are Graph Neural Networks?

Keyulu Xu, Weihua Hu, Jure Leskovec, Stefanie Jegelka.

In Proceedings of the 7th International Conference on Learning Representations (ICLR), 2019.

Oral Presentation (1.5% acceptance rate).

Representation Learning on Graphs with Jumping Knowledge Networks.

Keyulu Xu, Chengtao Li, Yonglong Tian, Tomohiro Sonobe, Ken-ichi Kawarabayashi, Stefanie Jegelka.

In Proceedings of the 35th International Conference on Machine Learning (ICML), 2018.

Long Talk (8% acceptance rate).

Distributional Adversarial Networks.

Chengtao Li, David Alvarez-Melis, Keyulu Xu, Stefanie Jegelka, Suvrit Sra.

In the 6th International Conference on Learning Representations Workshop Track (ICLR), 2018.

Generating Random Spanning Trees via Fast Matrix Multiplication.

Nicholas J.A. Harvey and Keyulu Xu.

In Proceedings of the 12th Latin American Theoretical Informatics Symposium (LATIN), 2016.

AWARDS AND
FELLOWSHIPS

Hudson River Trading AI Labs Fellowship, 2019

Chevron-MIT Energy Fellowship, 2019

David S. Y. and Harold Wong Fellowship, 2017

Andrew and Erna Viterbi Fellowship, 2016

Work Learn Research Award, 2014

Silver Medal, ACM-ICPC Programming Contest Pacific NW Region, 2013

INVITED TALKS

Microsoft Research, ML Ideas Seminar, New England, USA, April 2021

Foundations of Data Science Institute, ML for Algorithms, Cambridge, USA, April 2021

Imperial College London, MatchLab, London, UK, Feb 2021

Harvard University, Department of Mathematics, CMSA Seminar, Cambridge, USA, Nov 2020

Winter Festa Episode 5, Tokyo, Japan, Dec 2019

RIKEN AIP, Seminar, Tokyo, Japan, Aug 2019

Preferred Networks, Seminar, Tokyo, Japan, Aug 2019

Kyoto University, Kashima and Yamada Lab, Kyoto, Japan, Feb 2019

RIKEN AIP, Seminar, Tokyo, Japan, Aug 2018

PROFESSIONAL
SKILLS

Natural languages

Chinese, Japanese, English

Miscellaneous

CFA Level I Passed

PROFESSIONAL
SERVICE

Senior program committee

IJCAI 2021

Journal reviewing

IEEE Transactions on Pattern Analysis and Machine Intelligence

IEEE Transactions on Signal Processing

IEEE Transactions on Signal and Information Processing over Networks

IEEE Transactions on Evolutionary Computation

IEEE Transactions on Knowledge and Data Engineering

Conference reviewing

ICML (2020, 2021), NeurIPS (2020, 2021), ICLR (2021), AISTATS (2020, 2021)

COLT (2020), AAAI (2020, 2021), CVPR (2021), EMNLP (2021), ICCV (2021)